

A bit that flips at half period: a proof of Karttunen's conjecture on the binary columns of the powers of 3

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Abstract

Write the powers of 3 one under another in binary. The n -th column from the right is periodic in k with period dividing 2^n ; reading one such window as the coefficient vector of a polynomial over $\text{GF}(2)$ gives OEIS A037096. In January 1999 A. Karttunen conjectured that for $n \geq 3$ this polynomial is divisible by $(x + 1)^{3 \cdot 2^{n-2} - 1}$. We prove it, and show the exponent is exact. The whole content is a single 2-adic fact: $3^{2^{n-2}} \equiv 1 + 2^n \pmod{2^{n+1}}$, so advancing k by a quarter of the window *flips* bit n . A window built from a block, its complement, the block again and its complement again factors as $(x + 1)^{3q-1}(B(x)(x + 1) + x^q)$ with $q = 2^{n-2}$, and $3q - 1$ is exactly the conjectured exponent.

1 The sequence and the conjecture

OEIS A037096, contributed by Antti Karttunen on 29 January 1999 under the name *Periodic vertical binary vectors computed for powers of 3*, is

$$a(n) = \sum_{k=0}^{2^n-1} \left(\lfloor 3^k / 2^n \rfloor \bmod 2 \right) 2^k, \quad a(0), \dots, a(4) = 1, 2, 0, 204, 30840.$$

Identifying an integer with the $\text{GF}(2)$ polynomial whose coefficient of x^k is bit k , the entry carries

Conjecture: For $n \geq 3$, each term $a(n)$, when considered as a $\text{GF}(2)[X]$ polynomial, is divisible by the $\text{GF}(2)[X]$ polynomial $(x + 1)^{A055010(n-1)}$.

Here $A055010(m) = 3 \cdot 2^{m-1} - 1$ for $m > 0$, so the claimed exponent is $3 \cdot 2^{n-2} - 1$. As of the “Last modified” line on the live entry (23 August 2026) the statement is still recorded as an unproven conjecture and no proof appears in the entry's links or programs.

Throughout $n \geq 3$ is fixed and

$$L = 2^n, \quad q = 2^{n-2}, \quad b(k) = \lfloor 3^k / 2^n \rfloor \bmod 2, \quad A(x) = \sum_{k=0}^{L-1} b(k)x^k,$$

so A is the polynomial attached to $a(n)$. Note $L = 4q$. For a nonzero $P \in \text{GF}(2)[x]$ let $v(P)$ denote the exponent of $x + 1$ in P . We use $x^{2^m} + 1 = (x + 1)^{2^m}$, the Frobenius endomorphism in characteristic 2.

2 The 2-adic input

Lemma 1. *For every $m \geq 1$, $v_2(3^{2^m} - 1) = m + 2$. Consequently $3^{2^m} \equiv 1 + 2^{m+2} \pmod{2^{m+3}}$, and the multiplicative order of 3 modulo 2^{m+3} is 2^{m+1} .*

Proof. By the lifting-the-exponent lemma for $p = 2$ and even exponent N , $v_2(3^N - 1) = v_2(3 - 1) + v_2(3 + 1) + v_2(N) - 1 = 1 + 2 + v_2(N) - 1$. With $N = 2^m$ this is $m + 2$. Writing $3^{2^m} - 1 = 2^{m+2}u$ with u odd gives $3^{2^m} \equiv 1 + 2^{m+2} \pmod{2^{m+3}}$. The order statement is the standard consequence: $3^{2^m} \not\equiv 1$ and $3^{2^{m+1}} \equiv 1$ modulo 2^{m+3} . $\square$

Taking $m = n - 2 \geq 1$ and reading modulo 2^{n+1} :

$$3^q \equiv 1 + 2^n \pmod{2^{n+1}}, \quad q = 2^{n-2}, \quad (1)$$

and the order of 3 modulo 2^{n+1} is $2^{n-1} = 2q$, which divides L ; so b is well defined as a function on $\mathbb{Z}/L\mathbb{Z}$.

Lemma 2 (the flip). *For every k , $b(k + q) = 1 - b(k)$.*

Proof. Work modulo 2^{n+1} , which determines bit n . By (1),

$$3^{k+q} \equiv 3^k(1 + 2^n) = 3^k + 2^n 3^k \equiv 3^k + 2^n \pmod{2^{n+1}},$$

the last step because 3^k is odd, so $2^n 3^k \equiv 2^n$ modulo 2^{n+1} . Adding 2^n modulo 2^{n+1} complements bit n and changes nothing below it. $\square$

3 The theorem

Theorem 3. *Let $n \geq 3$, $q = 2^{n-2}$, and let $B(x) = \sum_{k=0}^{q-1} b(k)x^k$ be the first quarter of the window. Then in $\text{GF}(2)[x]$*

$$A(x) = (x + 1)^{3q-1} \left(B(x)(x + 1) + x^q \right),$$

and the second factor is not divisible by $x + 1$. Consequently

$$v(a(n)) = 3q - 1 = 3 \cdot 2^{n-2} - 1 = A055010(n - 1)$$

exactly, and in particular $(x + 1)^{A055010(n-1)}$ divides $a(n)$.

Proof. Write $S(x) = 1 + x + \dots + x^{q-1}$, the all-ones block of length q . By Lemma 2 the four consecutive blocks of length q making up the window are B , $B + S$, B , $B + S$ (complementing a 0/1 block is adding S over $\text{GF}(2)$; note $b(k + 2q) = b(k)$, consistent with the order being $2q$). Hence

$$A = B + x^q(B + S) + x^{2q}B + x^{3q}(B + S) = B(1 + x^q + x^{2q} + x^{3q}) + S(x^q + x^{3q}).$$

Now $1 + x^q + x^{2q} + x^{3q} = (1 + x^q)(1 + x^{2q})$ and $x^q + x^{3q} = x^q(1 + x^{2q})$, so

$$A = (1 + x^{2q}) \left(B(1 + x^q) + Sx^q \right).$$

In characteristic 2, $1 + x^q = (1 + x)^q$ and $1 + x^{2q} = (1 + x)^{2q}$ since q is a power of 2; and $S = (1 + x^q)/(1 + x) = (1 + x)^{q-1}$. Therefore

$$A = (1+x)^{2q} \left(B(1+x)^q + (1+x)^{q-1}x^q \right) = (1+x)^{2q}(1+x)^{q-1} \left(B(1+x) + x^q \right) = (1+x)^{3q-1} \left(B(x)(x+1) + x^q \right)$$

For the last claim, evaluate the cofactor at $x = 1$ over $\text{GF}(2)$: $B(1) \cdot (1 + 1) + 1^q = 0 + 1 = 1 \neq 0$, so $x + 1$ does not divide it. Hence $v(A) = 3q - 1$ exactly. $\square$

Remark (why $n = 2$ is excluded, and correctly so). Lemma 1 needs $m = n - 2 \geq 1$. At $n = 2$ the conclusion fails and so does the conjecture's hypothesis: $3^k \bmod 8 \in \{1, 3\}$, so bit 2 is never set, $a(2) = 0$, and there is nothing to factor. The entry's restriction to $n \geq 3$ is exactly the right one.

Remark (the sibling entry). A037096 records the identity $a(n) = A000215(n - 1) \cdot A037097(n)$ for $n \geq 2$, where $A000215(n - 1) = 2^{2^{n-1}} + 1$ is the Fermat number, i.e. $(x + 1)^{2^{n-1}}$ as a GF(2) polynomial. Theorem 3 splits off exactly that factor and leaves $(x + 1)^{2^{n-2}-1}(B(x)(x + 1) + x^q)$, so A037097 carries a conjecture of the same shape which should follow by the same argument under its own indexing. I have *not* checked A037097's definition or the status of its comment, and make no claim about it here.

Remark (honest assessment). The proof is five lines once Lemma 2 is in place, and Lemma 2 is lifting-the-exponent. Nothing here is deep; the natural destination is an OEIS comment. What kept the statement open for twenty-seven years is presumably that the object is homemade and nobody looked. The same mechanism — a structural symmetry of the bit column turning into a polynomial factor — settles the companion conjectures on A036284, treated elsewhere; there the symmetry is a halved period rather than a flip.

4 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) recomputes $a(0), \dots, a(8)$ from the defining sum and matches the published DATA exactly; (ii) confirms that the multiplicative order of 3 modulo 2^{n+1} is 2^{n-1} and that $3^{2^{n-2}} \equiv 1 + 2^n \pmod{2^{n+1}}$ for $3 \leq n \leq 24$; (iii) confirms Lemma 2 at every k for $3 \leq n \leq 15$; (iv) confirms the factorization of Theorem 3 by explicit polynomial multiplication for $3 \leq n \leq 14$; and (v) computes $v(a(n))$ for $3 \leq n \leq 14$, obtaining 5, 11, 23, 47, 95, 191, 383, 767, 1535, 3071, 6143, 12287, which is $3 \cdot 2^{n-2} - 1$ in every case, together with cofactor value 1 at $x = 1$ in every case, confirming exactness.

References

- [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A037096>, entry A037096, consulted 23 August 2026; also A055010 and A036284.
- [2] P. Ribenboim, *The New Book of Prime Number Records*, 3rd ed., Springer, 1996 (for the lifting-the-exponent lemma at $p = 2$).

A Suggested OEIS comment (plain text, house style)

The conjecture is true, and the exponent is exact. Fix $n \geq 3$, put $q = 2^{n-2}$, and let $b(k)$ be bit n of 3^k . By lifting the exponent, $v_2(3^{2^m} - 1) = m + 2$, so with $m = n-2$ we get $3^q \equiv 1 + 2^n \pmod{2^{n+1}}$. Since 3^k is odd, $3^{k+q} \equiv 3^k + 2^n \pmod{2^{n+1}}$, and adding 2^n modulo 2^{n+1} complements bit n . Hence $b(k+q) = 1 - b(k)$: advancing k by a quarter of the window flips the bit. The order of $3 \pmod{2^{n+1}}$ is 2^{n-1} , so the window of length $2^n = 4q$ consists of a block B , its complement,

B again, and its complement again. Over $GF(2)$, complementing a block of length q is adding $S = 1 + x + \dots + x^{(q-1)} = (x+1)^{(q-1)}$. Therefore

$$\begin{aligned} a(n) &= B(1 + x^q + x^{2q} + x^{3q}) + S(x^q + x^{3q}) \\ &= (1 + x^{2q})(B(1 + x^q) + S x^q) \\ &= (x+1)^{(2q)} * (x+1)^{(q-1)} * (B(x)(x+1) + x^q) \\ &= (x+1)^{(3q-1)} * (B(x)(x+1) + x^q), \end{aligned}$$

using $1 + x^{(2^j)} = (x+1)^{(2^j)}$ in characteristic 2. Here $3q - 1 = 3 \cdot 2^{(n-2)} - 1 = A055010(n-1)$. The cofactor evaluated at $x = 1$ is $B(1) \cdot 0 + 1 = 1$, so $x+1$ does not divide it and the exponent is exactly $A055010(n-1)$, not merely at least it. For $n = 2$ the argument does not apply and indeed $a(2) = 0$, since $3^k \bmod 8$ is 1 or 3 and bit 2 is never set.